BodyCheck

AI-Powered Medical Triage & 3D Symptom Checker

Team Clavicular · HCMUS APCS/CLC · LotusHacks 2026

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1 Feasibility Assessment

This section addresses the five feasibility sub-criteria defined in the Vietnamese Student HackAIthon 2026 Bang B (Challenger) Round 1 rubric (*Nhom tieu chi 3: Tinh kha thi* — 25 points): legal data sourcing and qualified personnel; technical build and deployment viability; infrastructure and operational cost estimation; security and legal compliance under current Vietnamese law; and the post-competition GTM roadmap. Evidence is drawn from the team's direct engineering experience and the organiser-provided VNPT API ecosystem that will power the Round 2 build.

Table 1 gives a structured overview; each dimension is examined in depth in the subsections that follow.

Table 1: BodyCheck five-axis feasibility assessment aligned to the HackAIthon 2026 Bang B Round 1 rubric (25 pts total). Risk ratings: **Low** = controls in place; **Medium** = mitigations identified.

Dimension	Key Evidence	Risk	Current Status
Legal Data & HR	No PHI stored; VNPT APIs under organiser-provided access; 4-person HCMUS CS student team	**Low**	Cleared for Round 1 proposal
Tech Build & Deploy	Three-tier Flutter / Node.js / FastAPI stack; VNPT API integration path confirmed	**Low**	Round 2 MVP deliverable within 8 days
Infra & Costs	VNPT APIs free in Round 2 (organiser-provisioned); single low-cost compute instance post-competition	**Medium**	Post-competition billing TBD†
Security & Legal	TLS enforced; vault secrets; AI disclaimer embedded; general info tool classification at MVP stage	**Medium**	Clinical/legal review before public release
Roadmap & GTM	Three-phase post-competition plan; B2C + B2B2C channels; 12-month market expansion roadmap	**Low**	Detailed in §1.6

†Post-competition API pricing and compute cost estimates will be confirmed once commercial VNPT tier information is available (see `QUESTIONS_for_team.md`).

1.1 Legal Data Acquisition and Human Resources

Data Sourcing and Privacy Compliance

BodyCheck is architected to avoid handling protected health information (PHI) at the infrastructure level. Symptom descriptions entered by users are treated as transient, request-scoped payloads: they are forwarded to the AI reasoning layer, a structured clinical summary is generated, and the raw input is discarded without being written to any persistent store or log file in identifiable form.

Medical reference content is retrieved at inference time from authoritative public-facing sources via API rather than ingested into a proprietary dataset. In Round 2 of the competition, this retrieval layer will be implemented using **VNPT Smartbot** (advanced LLM-backed question answering, API 4.2) and **VNPT SmartReader** (document OCR and information extraction, API 5.1/5.2), both provided by the organiser under the competition's API access agreement. This eliminates all data-licensing risk for the competition phase: every API call is governed by VNPT's enterprise terms and is explicitly sanctioned for participating teams by the organiser (*Ban To chuc*).

The system therefore operates without requiring independent data-collection agreements, proprietary patient datasets, or any form of model fine-tuning on personally identifiable health records.

Team Composition and Skill Coverage

The team consists of four full-time undergraduate students from the Advanced Program in Computer Science (APCS) and the High-Quality Program (CLC) of the Faculty of Information Technology, Ho Chi Minh City University of Science (HCMUS). Each member holds unambiguous ownership of a distinct engineering domain:

- **AI Engineering** (*Nguyen Dinh Thien Loc*): LLM orchestration, VNPT Smartbot and SmartVoice integration, FastAPI AI microservice design, and prompt engineering.
- **Frontend / UI** (*Quach Thien Lac*): Flutter mobile/web client, Three.js 3D anatomical canvas (iframe wrapper), and VNPT SmartUX analytics embedding.
- **3D Asset Design** (*Nguyen Quoc Nam*): Blender modelling, GLB export and optimisation, and clickable anatomical region mapping across three model files.
- **Backend / Gateway** (*Tran Ton Minh Ky*): Node.js/Express API gateway, JSON schema validation middleware, VNPT eKYC integration, and error-handling controllers.

All four members are currently enrolled full-time students, satisfying the eligibility criteria for Bang B – Challenger (up to five members per team, including an optional faculty advisor). The team is self-sufficient at the MVP scale required for Rounds 1 and 2. Post-competition scaling toward a production-grade platform would additionally require a DevOps/SRE engineer and a licensed medical content adviser.

As a validation point, the team built and demonstrated a working first prototype of the core triage loop — 3D body-part selection, symptom input, AI-generated clinical summary, and clinic geo-routing — in a 72-hour sprint at a prior student hackathon, confirming that the stack is viable and that the team can deliver under time pressure.

1.2 Technical Build and Deployment Feasibility

Technology Stack Viability

BodyCheck employs a *decoupled three-tier service architecture* that separates public-facing API traffic from the AI orchestration layer, making each tier independently testable, replaceable, and deployable:

- **Flutter Web/Mobile (Client Tier):** A single Dart codebase compiles to native mobile (ARM) and optimised web (Wasm/JS), so the same product ships as a PWA and as a native app without a separate codebase. The interactive 3D anatomical model is rendered by a Three.js scene embedded as an iframe; region-selection events travel back to Flutter via `postMessage`.
- **Node.js 20 LTS / Express 5 (Gateway Tier):** Handles CORS, JWT authentication, JSON Schema request validation, and reverse-proxying to the AI microservice. Node's event-loop model absorbs concurrent upstream API calls without thread contention.
- **Python 3.12 / FastAPI (AI Orchestration Tier):** ASGI-native, Pydantic-validated. FastAPI provides auto-generated OpenAPI docs, sub-millisecond request parsing, and clean async bindings for all VNPT client SDKs.
- **VNPT API Ecosystem (Round 2):** The organiser provides direct access to eight API families. BodyCheck's planned integration uses: SmartVoice STT (API 3.2) for voice symptom input; SmartVoice TTS (API 3.1) for reading responses back to users with accessibility needs; SmartVoice call-summary (API 3.3) for distilling multi-turn consultation transcripts into a structured summary; Smartbot rule-based triage (API 4.1) for fast, deterministic first-pass classification; Smartbot advanced LLM (API 4.2) for open-ended differential reasoning and follow-up chat; SmartReader OCR and extraction (API 5.1/5.2) for parsing user-uploaded medical documents, lab reports, and prescriptions; SmartUX analytics (API 7.1) for in-app interaction telemetry and friction-point identification; vnSocial trend listening (API 6.1/6.2) for monitoring health-topic sentiment on Vietnamese social media, surfacing epidemic signals and common symptom clusters; eKYC document OCR (API 1.1) for verified account creation in the B2B2C institutional channel; and SmartVision face detection (API 8.3) as an optional telemedicine extension that confirms user presence before a live consultation handoff. Using the full breadth of the organiser-provided API catalogue maximises the integration score in the Round 2 product completion rubric and demonstrates the depth of the team's engagement with the VNPT ecosystem.

Figure 1 shows the complete deployment topology, including the VNPT API integration layer.

Local Development and CI/CD Workflow

All three tiers run concurrently on a single developer laptop: Uvicorn on port 8016, Express on port 3000, and `flutter run -d chrome` pointing at the local gateway. A new contributor can clone the repository and have every service running in under 15 minutes.

For the Round 2 submission, each service is containerised with a minimal `Dockerfile` and composed via `docker-compose.yml`, satisfying the competition's one-command setup requirement for product completion. A lightweight GitHub Actions workflow runs automated

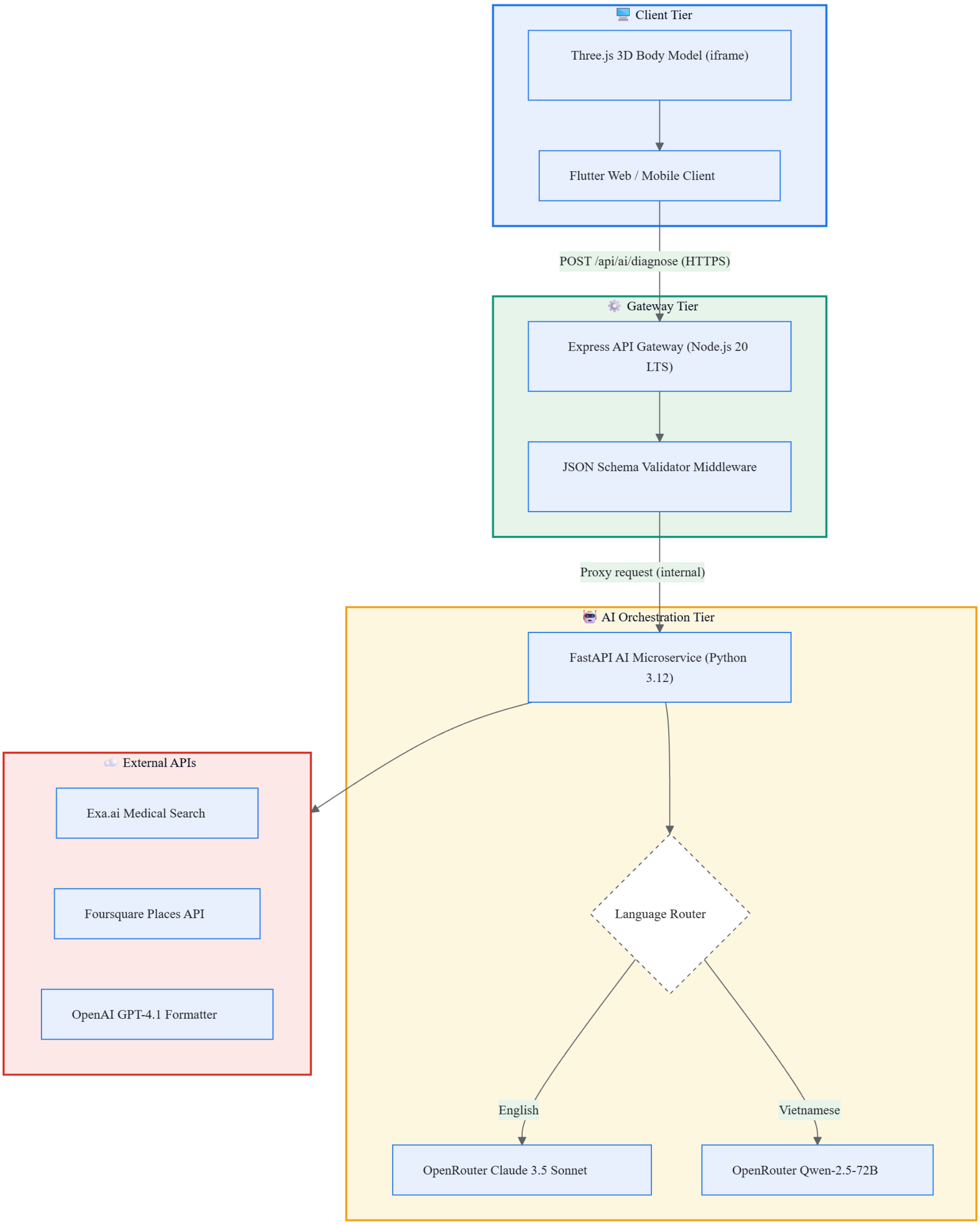

Figure 1: BodyCheck three-tier deployment topology with VNPT API integration. Arrows denote synchronous HTTPS calls between components; shaded subgraphs are independently containerisable service tiers. The AI Orchestration Tier calls VNPT SmartVoice, Smartbot, and SmartReader in Round 2; post-competition builds may extend to additional partner APIs.

smoke tests against the `/api/diagnose`, `/api/chat`, and `/api/clinics` endpoints before each push to the submission branch.

1.3 Infrastructure and Operational Costs

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Round 2 — Competition Phase

During Round 2 (26 June – 3 July 2026), the organiser provides all VNPT API access at no cost to the team. Infrastructure expenditure is therefore limited to the compute instance required to run the three-tier service stack — a university lab server or a minimal cloud VM (2 vCPU, 2 GB RAM) is sufficient for the prototype load expected during judging and live-demo sessions.

Post-Competition Cost Model

After the competition, the team must transition to a commercial or partner API pricing arrangement. The total monthly operational expenditure scales linearly with session volume R:

$$C_{\text{ops}}(R) \;=\; C_{\text{fixed}} \;+\; \sum_{i=1}^{n} c_i\,\alpha_i\,R \tag{1}$$

where C_{fixed} covers compute, networking, and storage; c_i is the per-call unit price of VNPT service s_i; and α_i is the number of calls per session. Because the AI token budget per session is bounded by a fixed prompt template, this cost model is *strictly linear* in R with no quadratic blowup.

The break-even monthly session volume R^* under monetisation price p satisfies:

$$R^* \;=\; \frac{C_{\text{fixed}}}{p \;-\; \sum_{i=1}^{n} c_i\,\alpha_i} \tag{2}$$

Note on specific figures: Commercial per-call pricing for VNPT API tiers is not yet confirmed. Detailed cost estimates will be inserted once the team establishes a commercial arrangement with VNPT or an equivalent partner post-competition. The structural model above remains valid regardless of the specific coefficient values.

1.5 Security and Legal Compliance

Data Protection Architecture

BodyCheck treats all symptom inputs as potentially sensitive health information, applying defence-in-depth controls at every tier:

- **Transport Security:** All client-to-gateway and inter-service communication uses TLS 1.3. Internal service calls within a private VPC stay off the public internet.
- **Secrets Management:** API credentials (VNPT keys, gateway tokens) are injected as environment variables from a secrets vault (Docker Secrets or equivalent) and never appear in source control. The repository includes only a `.env.example` template listing required variable names without exposing values.

- **Input Validation:** The Express gateway applies JSON Schema middleware before any payload reaches the AI tier, blocking prompt-injection and malformed-input vectors at the network boundary.
- **No Persistent PHI:** The FastAPI service holds session context only within the lifespan of a single HTTP request. Structured logs record only anonymised metadata (request ID, latency, model tag, HTTP status). No free-text symptom content appears in log output.
- **Rate Limiting:** Per-IP rate limits and API-key quotas at the gateway prevent runaway API quota consumption and abuse of the organiser-provided VNPT allocation.

Open-Source Licensing and Medical Liability

The BodyCheck codebase is released under the MIT License, which explicitly disclaims all warranties and limits liability. Because the system produces health-adjacent guidance, additional framing is mandatory:

- A medical disclaimer is embedded in every AI response payload and displayed prominently in the Flutter UI, clearly stating that BodyCheck provides *general informational support only* and is not a substitute for professional medical advice.
- Emergency escalation text (Vietnam emergency line: **115**) is injected by the prompt template whenever the AI detects a high-severity symptom pattern.
- Pre-launch clinical and legal review by a licensed medical professional and by legal counsel specialising in Vietnamese health-sector regulations is required before any public-facing deployment beyond the competition context.

Regulatory Classification

Under current Vietnamese law, the relevant authorities are the Ministry of Information and Communications (MOIT) for digital platform registration and the Ministry of Health (MOH) for services classified as medical devices or telemedicine under Circular 30/2020/TT-BYT.

At MVP stage, BodyCheck is a *general health information tool*, not a telemedicine service or medical device, provided the non-diagnostic framing and mandatory disclaimer are consistently maintained. This keeps the competition-phase product outside medical device registration requirements. Future versions targeting structured diagnostic conclusions would require registration under Decree 98/2021/ND-CP and may require ISO 13485 quality management certification.

1.6 Post-Competition Roadmap and GTM Strategy

BodyCheck's market scope is intentionally *global from day one*: the entire product interface, AI prompt layer, and clinical-reference content operate in English, making it accessible to users in any country without a localisation bottleneck. Vietnam serves as the primary launch market given the team's local regulatory familiarity and existing network, with South-East Asia and broader English-speaking markets as the natural second wave.

Beyond the primary consumer health use case, BodyCheck addresses a material problem in **education**: medical and nursing students, first-aid training programmes, and health-literacy curricula routinely struggle to teach anatomical symptom recognition interactively. The same 3D body-region interface and triage-reasoning engine that serves a self-diagnosing user also serves a student learning clinical decision pathways — creating a dual-market opportunity from a single codebase. This educational channel aligns directly with HackAIthon topic 1 (AI solutions supporting Personalised Learning) in addition to the primary topic 4 (AI supporting Healthcare Operations).

The post-competition roadmap is structured in three phases aligned with the rubric's 12-month market expansion requirement.

Phase Definitions

Phase 1 — HackAIthon Competition

June – July 2026 (Rounds 1–3).

Submit the Round 1 idea proposal. If selected for Round 2 (announcement before 26 June), integrate as many VNPT API services as possible into the BodyCheck stack over the 8-day build window (26 June – 3 July) with direct Mentor support. Demonstrate a live-running product at the Round 3 finals (14–15 July 2026 in Hanoi). Collect structured judge and mentor feedback to anchor the post-competition product direction.

Phase 2 — Hardening and Closed Beta

August – November 2026 (Months 1–4 post-competition).

Begin serious post-competition development in August once semester schedules allow. Stabilise and document all VNPT API integrations; containerise all services with a one-command `docker-compose` setup; implement TLS certificate provisioning and vault-based secret injection. In September–October, run a closed beta with 30–50 volunteer testers drawn from HCMUS student and faculty networks and online medical communities. Iterate on VNPT Smartbot prompt templates and SmartReader parsing pipelines based on real-user feedback. Conduct a first-pass clinical language and legal review. Target: a stable, security-hardened beta release by end of November 2026.

Phase 3 — Public Launch and Market Expansion

December 2026 – June 2027 (Months 5–12 post-competition).

Add persistent session storage, user account management, and a UX metrics observability stack (structured logging, distributed tracing). Launch the consumer app (B2C) as a PWA and on mobile stores with a freemium model in February 2027. In parallel, onboard the first education-sector partners (medical schools, online health-literacy platforms) under a white-label API licence in March 2027, followed by the broader B2B2C healthcare channel (telehealth operators, hospital intake portals) in April 2027. Pursue institutional funding via accelerator programmes (VNPT Innovation Hub, Google for Startups Vietnam, HCMUS technology-transfer office) starting January 2027. Initiate MOH regulatory consultation for the clinical-grade product tier in April 2027. Target: stable public launch **v1.0** by June 2027, approximately 12 months post-competition.

Go-To-Market Channel Strategy

BodyCheck targets three complementary GTM channels:

1. **B2C — Global Consumer Health App (English-first):** A free-tier PWA and native mobile app distributed worldwide. The English-first UI removes localisation barriers and makes the product immediately accessible across South-East Asia, South

Asia, and other regions underserved by AI-powered medical triage tools. Revenue via freemium subscription.

2. **B2B2C — API Licensing to Healthcare Platforms:** White-label REST API for telehealth operators, hospital patient intake portals, and health insurance triage systems. Priced at a per-call rate plus a monthly platform fee. Primary targets are Vietnamese healthcare organisations; secondary targets are international digital-health platforms that need a multi-language AI triage layer.

3. **Education Channel — Medical Training Platforms:** The same 3D anatomical model and AI reasoning engine that supports self-triage also serves medical and nursing students, first-aid training programmes, and health-literacy curricula. This channel is licensed to universities and online learning platforms as a white-label simulation tool, at a flat institutional annual fee rather than per-session billing.

Figure 2 shows the full 12-month roadmap anchored to the competition milestone.

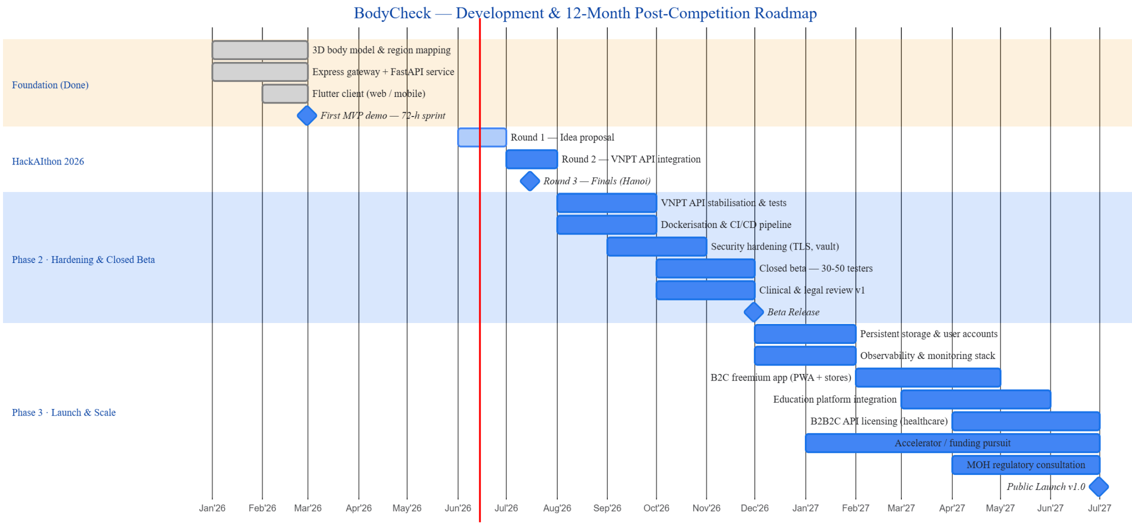

Figure 2: BodyCheck 12-month development and GTM roadmap anchored at the HackAIthon 2026 finals (July 2026). Phase 1 covers the competition period; Phases 2 and 3 represent the post-competition commercialisation trajectory. Completed tasks appear in green; planned tasks in blue; diamonds mark major milestones.

Table 2: Principal risks and mitigation strategies for the post-competition product roadmap.

Risk	Severity	Mitigation Strategy
VNPT commercial API pricing	Medium	Provider-agnostic routing layer in FastAPI allows hot-swapping to alternative AI backends (e.g., open-source LLM hosts, Gemini API) without touching application code above the service boundary.
Regulatory reclassification	Medium	Phase 3 launch targets the non-regulated consumer information category. Clinical-grade features are deferred behind a separate product tier pending MOH review. The mandatory disclaimer ensures compliance throughout.
Team continuity	Medium	Infrastructure-as-Code, clean API boundaries between tiers, and comprehensive technical documentation lower the barrier for new contributors and reduce single-point-of-failure risk.
User trust in AI medical tools	Medium	Transparent uncertainty communication, mandatory disclaimer, and triage-support framing (not diagnosis) are embedded in the prompt template and UI copy at every touchpoint.